

VR-enabled intelligent space planning platform for facility layout redesign

Zhengji Liang

Siu Fung Wong

Leith Kin Yip Chan*

Zixuan He

Department of Industrial and Manufacturing Systems Engineering

The University of Hong Kong

ABSTRACT

The facility environment is a workspace for employees to engage in creative endeavours. However, due to the dynamic nature of human resources, the interior layout of the facility often requires frequent adjustments. Typically, the facility's interior environment is relatively fixed because of the stable equipment, allowing limited changes during the space planning process. The challenge lies in achieving maximum available working space while minimising alterations to the layout. To address this problem, this paper proposes a framework of intelligent space planning platform for facility layout redesign projects, leveraging VR technology and optimisation algorithms within a simulated virtual world. Users can compare different layout design alternatives, optimise equipment allocation, and design furniture placement routes. Ultimately, the platform aims to provide the optimal layout plan and an optimised in-house logistics plan for the renovation process. Through this method, this research contributes to the improvement of functionality, safety and efficiency in facility environments.

Keywords: Virtual Reality, Space planning, Facility layout plan.

Index Terms: Human-centered computing Scenario-based design; Human-centered computing Virtual reality; Software and its engineering Virtual worlds training simulations

1 INTRODUCTION

The office environment often becomes a second home for many individuals, as they spend decades dedicated to their careers to sustain their livelihoods. Similarly, in educational institutions, laboratories serve as essential spaces for students to study and teachers to work. Creating a conducive and comfortable co-working environment within these facilities can significantly enhance employee satisfaction. Such an environment has the potential to boost both efficiency and creativity, fostering a more productive and innovative workspace.

Because of dynamic changes in human resources and facility equipment, there always exists a space organisation problem: how can we arrange more available working space and tidy up the indoor workspace? It has been proved that in the design phase, Virtual Reality (VR) is helpful to visualise the design alternatives [1]. While in the maintenance phase, can VR help with space reorganisation? To answer this research question, this paper intends to investigate the application of VR technology in an intelligent space design platform for facility layout redesign and implement optimisation algorithms to enhance the intelligence of the internal space renewal process. The objective of this project is to utilise VR technology to redesign the facility space, thereby improving its functionality, safety, and efficiency.

*email: lkychan@hku.hk

2 LITERATURE REVIEW

To answer the research question mentioned above, this session aims to review pertinent literature in two domains: VR application in interior design and space optimisation theory within facility environments.

2.1 VR application in interior design

Since the boosting development of VR technology, it has arisen research interests in its integration with industrial application scenarios. VR distinguishes itself from other forms of human-computer interfaces by actively engaging its user within the virtual environment rather than employing it as a passive user [2]. Because of the unique advantage of offering the user with a sense of space, VR has primarily been used as a visualisation tool in architectural design [3]. Despite the role of visualisation tools, VR can function as a collaboration platform. Du developed a multiplayer cooperation platform in VR named CoVR, which was expected to help project stakeholders create a shared vision before construction work commences. The initial results of their research indicated that it could improve interpersonal interactions in the immersive virtual environment and enhance communications in the construction tasks [4]. However, there exists a research gap in its application as a decision-making assistant within the realm of 3D visualisation. The potential scenarios where VR could serve as such an assistant have yet to be fully explored. Therefore, this research intends to fill the research gap by developing an intelligent space planning platform, especially for facility layout redesign projects.

2.2 Space optimisation in the facility

Generally, the co-working space users are freelancers and self-employed workers. However, the user groups can be extended to firms in small or large size and students. The core values behind co-working space are collaboration, community, accessibility, sustainability and openness. The general physical design of co-working is an open floor plan with shared workspaces where the co-workers attach importance to its atmosphere and layout [5]. Integrated with advanced data-driven methods, Carl developed a set of procedural algorithms for space planning in commercial offices, which can be performed as an architect and improve the efficiency and effectiveness of the design process [6]. Tanasra applied a machine learning tool to automatically create a furniture layout, which can be used for enhancing the design process [7].

It has been proved that layout is one of the key features in the relationship [8]. In layout design, ergonomics considerations can facilitate occupational safety and health, work efficiency and employee well-being [9]. A well-designed interior office space can create an atmosphere where the employee feels comfortable. Therefore, their creativity will be inspired. Visual harmony of the interior space can be generated by defining colour, textures, forms and empty space [10]. In the digitalisation era, designers can conduct simulation experiments using digital human models in a virtual environment to assess the ergonomic comfort of interior layouts and furniture design [11]. Inspired by the above research, this paper intends to extend the application of VR beyond

ergonomics design to include space design simulations for the facility layout redesign.

3 VR-AIDED SPACE DESIGN METHODOLOGY

Learned from the literature study, this section presents the systematic methodology design, encompassing data collection in the physical environment, digital transformation procedure, and collaboration platform development in the virtual environment. Firstly, special data will be collected using Point Cloud technology from the physical environment. Subsequently, through the software's digital transformation process, layout plans and 3D models of furniture accessories will be generated, forming the basis for constructing a VR scene representing the physical workspace in the virtual environment. The proposed intelligent space design platform will obtain input data from the interactive collaboration platform within the virtual environment. With the help of optimisation algorithms and in-house renovation logistics simulation, the outputs of the system will be the optimal space redesign pan and optimised in-house logistics plan. Fig. 1 illustrates the technical route.

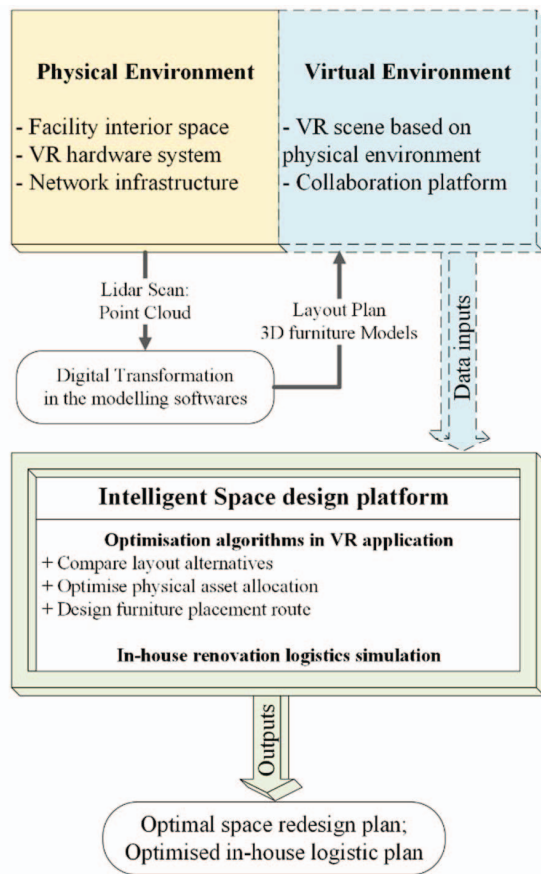


Figure 1: Technical route of the project

3.1 Data collection in the physical environment

The physical environment considered in this research is defined as basic equipment and the corresponding building environment. Fig. 1 provides an overview of the three key elements comprising the physical environment: the facility interior space, the VR hardware system, and the network infrastructure. The facility's interior space accommodates the VR system and associated furniture. The

network infrastructure plays a crucial role in facilitating the operation of the VR system.

To collect thorough geometry data of the interior environment, Point Cloud technology is designed to be applied in the data collection process. The raw data of the facility's interior environment was generated from Lidar Scanning. Fig.2 shows the panoramic photo generated from the scanned facility's interior environment using Point Cloud technology.



Figure 2: Panoramic picture generated from Point Cloud

3.2 Digital transformation procedure

After the necessary data collection process, raw data will be transformed into 3D furniture models and an original layout plan through software. The HKU Human-systems Interaction and Simulation (HIS) Laboratory was scanned using LiDAR technology as a case in this research. The resulting data was processed in the FARO SCENE [12] to generate a point cloud. This point cloud was then converted into a triangle mesh, which can be displayed in the Cave Automatic Virtual Environment (CAVE) system using the Extended Reality Content Creator (XRCC). The XRCC can provide a cost-effective and user-friendly VR content creation tool for domain-specific training and operational scenarios [13]. However, the high polygon count of the model can negatively impact performance and user experience in Virtual Reality. To address this issue, there are methods to filter out stray points from the point cloud and decimate the complexity of the triangle mesh. However, these methods can be computationally intensive and may decrease accuracy.

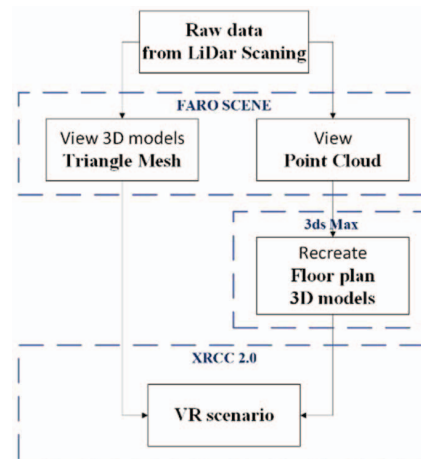


Figure 3: Digital transformation process

The expected rendering result is displayed in Fig. 4. The digital transformation process at this stage is shown in Fig. 3. Meanwhile, a deep learning model for segmentation and classification of point cloud data is planned for the next stage of the research. The foundational architecture of this model draws inspiration from

PointNet++, a hierarchical neural network characterised by its multilayered structure. This design allows for the effective extraction of prominent features from 3D scenes, enhancing the model's interpretative capabilities [15]. For data acquisition, we will utilise datasets from ModelNet 40 [16] and Redwood [17], which include coloured 3D indoor data. The diversity of these datasets is critical for training and testing the model, ensuring high accuracy across various scenarios.

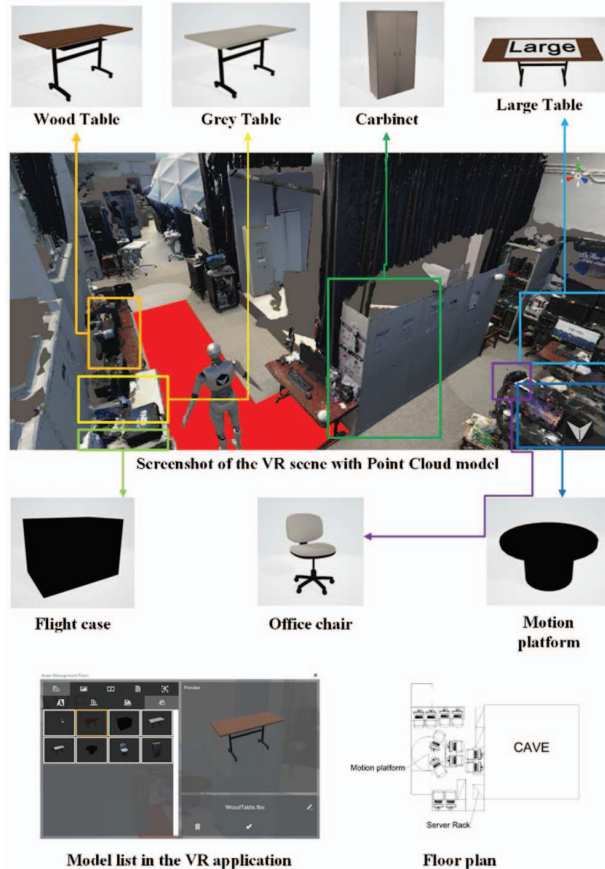


Figure 4: Digital transformation results

3.3 Collaboration platform development in the virtual environment

Apart from the traditional presentation of space plan proposals, this research endeavours to enhance the intuitive display of layouts. The incorporation of VR technology enables an immersive environment, offering users the opportunity to visualise 1:1 scale models from a 3D perspective, thereby fostering a better understanding of the design plan.

To discuss the feasibility, the layout alternatives will be included in the scenarios. As shown in Fig. 5, a scene menu with the screenshot of the layout plan will be developed in the user interface to facilitate the decision-making process. By clicking the photos of the layout plan in the menu, the user can switch between scenes, compare alternatives, and express their preferences. For designers, the VR application facilitates the demonstration of their proposed space, enabling them to effectively convey their design concepts and receive real-time feedback from the audience. In this manner, VR serves as a communication tool, enhancing the interaction and collaboration between designers and stakeholders.

3.4 In-house renovation logistics simulation

The intelligent space design platform is designed to provide in-house renovation simulation. This research mainly focuses on furniture replacement in the facility layout redesign process. By utilising the VR application, users will gain an initial perspective on how to allocate furniture when modifying the layout. The algorithm plug-in will calculate the optimal furniture placement route based on the layout plan, subsequently generating an animated navigation process automatically. This integrated system aims to streamline and enhance the furniture allocation process, contributing to a more efficient and effective facility layout redesign.



Figure 5: User interface design in the application

4 SPACE DESIGN OPTIMISATION ALGORITHM

In this part, optimisation algorithms for space design will be introduced. Since this project is a work-in-progress, the introduction only covers the design concept of the algorithms. These algorithms are converted from user needs and general principles for interior space design. There are three fundamental algorithms: layout alternative comparison algorithm, physical asset allocation optimisation algorithm, and furniture placement route calculation algorithm. They will be implemented as plug-ins into the intelligent space plan platform.

4.1 Layout alternatives comparison

To compare the design alternatives of the facility layout, this research converted the user requirements to the design criteria. According to the interviews with the employees from the initial case study in HKU, the general needs are summarised as functionality, safety and efficiency. Based on the basic requirements mentioned before, the criteria will be the total available working area and the average access distance to the facilities. By comparing the key criteria, an optimal layout plan will be voted as a final decision.

The total available working area will be calculated from the personal workstation area and the overlap area in the room. The standard of private workstation area will be designed from ergonomics data and specific requirements. Therefore, in this research, the definition is the area where each person has their own activity area, and a set of equipment is located (computer, keyboard, mouse, etc.). Inspired from offset geometry proposed by Schwabe et al. [18], this research will introduce the concept of overlap area. The overlap area will be considered as the conflict area where the personal workstations overlap or the workstation conflicts with the access area of the furniture. The average access distance is responsible for the average distance of all the access distance between each worker and the target facility.

The calculation criteria assumptions are as follows:

- Sp : private workstation area per person

- $S_{overlap}$: total overlap area
- TS : total available working area
- D_{ij} : the access distance of the i^{th} object from the j^{th} worker
- D_{avg_i} : the average access distance of the i^{th} object
- D_{avg} : the average access distance of all the objects
- m : the number of workers
- n : the number of objects

The equation of key criteria calculation as follows:

$$TS = n * Sp - S_{overlap} \quad (1)$$

$$D_{avg_i} = \frac{\sum_{j=1}^m D_{ij}}{m} \quad (2)$$

where $j = 1, 2, \dots, m$

$$D_{avg} = \frac{\sum_{i=1}^n D_{avg_i}}{n} \quad (3)$$

where $i = 1, 2, \dots, n$

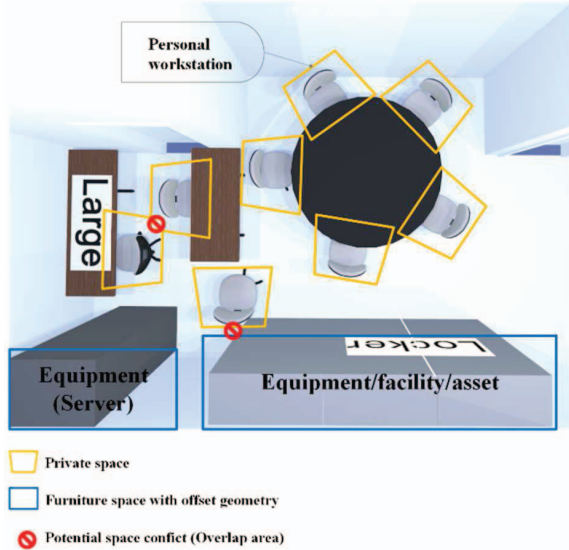


Figure 6: Available area calculation assumptions of a layout plan

The two objectives of the comparison are to find an optimal layout that satisfies values of the total available working area and the average facility access distance. A larger value for the total available working area is considered favourable, indicating a more spacious and functional layout. Conversely, a smaller value for the average facility access distance is deemed more optimised, as it signifies shorter distances to reach essential facilities within the layout.

4.2 Physical asset allocation optimisation

In the facility environment, physical asset in this research is defined as the physical tools used in daily operation, such as necessary equipment, computer accessories, cables, etc. The physical asset will be labelled with the frequency of usage and its volume. The golden rule of allocation is that the distance between equipment and its user should be determined by its frequency of usage. The physical asset allocation optimisation algorithm aims to

answer the question of how to organise different sizes of physical assets. Spatial constraints are simulated using a physical system within a game engine, effectively replicating the real-world limitations and boundaries that influence the arrangement of assets within the laboratory environment.

The vectors of the physical asset in the allocation process are described in the following:

- L : the length of the storage space
- W : the width of the storage space
- H : the height of the storage space
- V : the volume of the storage space
- L_i : the length of the object i
- W_i : the width of the object i
- H_i : the height of the object i
- V_i : the allocated storage space of the object i

The objective is to calculate the allocation combination which can remain the maximum storage space after putting in all the assets in the storage position. The equation is shown below:

$$Obj. \text{ Max } f(V_i, V) = V - \sum_{i=1}^n V_i \quad (4)$$

where $i = 1, 2, \dots, n$

$$Subject \text{ to } \sum L_i \leq L, \sum W_i \leq W, \sum H_i \leq H \quad (5)$$

The algorithm will solve the spatial arrangement problem in the simulated virtual world. Its output provides an optimized location for physical assets, ensuring maximum remaining space for users to accommodate additional items. Furthermore, the algorithm considers usage habits and strategically places the most commonly used items on the outer layer for easy access. By optimising the arrangement of physical assets, the algorithm enhances both space utilisation and user convenience.

4.3 Furniture placement route design

When the location of the equipment has been confirmed, the subsequent challenge lies in correctly positioning them. After finalising the allocation of individuals and assets, the design of a furniture placement route becomes crucial to achieve the desired layout. The spatial constraint here involves considering the correct orientation and articulation of the furniture's position.

In the process of route planning, several practical problems arise. Firstly, it is essential to avoid collisions between the furniture being transported from its original position to the target destination and any existing objects in the path. Secondly, minimising the total distance travelled by the furniture during transportation is another key consideration. For instance, within the VR development software, furniture may be recognised as "game objectives," transforming the route design problem in the game engine into a navigation problem for these objects.

To prevent collisions, a collision detection mechanism can be implemented where the object adjusts its course when it comes into contact with the collider of another nearby object. This adjustment may involve turning at a certain angle to maintain a safe distance and avoid any potential clashes. By addressing these practical challenges in route planning, the optimised placement of furniture can be achieved while ensuring efficient navigation and minimising any potential disruptions.

5 CONCLUSION

This research addresses the challenge of dynamic interior layout changes in facility interior environments by proposing a framework of the intelligent space planning platform. By harnessing the power of VR technology within a simulated virtual world, the platform enables users to compare various layout design alternatives based on available working space and average facility access distance and devise efficient furniture placement routes after optimising physical asset allocation position. Through this approach, the platform aims to provide the optimal layout plan and an optimised in-house logistics plan for facility renovation projects.

By utilising VR technology as a decision-making assistant in the space planning process, the intelligent platform offers a novel and immersive way to visualise and redesign facility working spaces. This not only enhances the functionality and efficiency of the workspace but also provides users with a better understanding of the design concepts. The platform's ability to dynamically adjust the interior layout while maximising available working space minimises disruptions and ensures a seamless transition during the renovation process. Overall, this research contributes to the advancement of intelligent space design and optimisation in facility environments.

The limitation of this research pertains to the current reliance on manual design processes. At present, the modelling phase is carried out by an expert in the laboratory, which may not be feasible or available in other facility settings. Consequently, future iterations of the project aim to address this limitation by incorporating machine learning techniques into the design evaluation and optimisation process. For example, a deep learning model for segmentation and classification of point cloud data will be implemented in the data processing phase. This model will be engineered to streamline the VR scene modelling process, enabling the automatic identification of elements within the environment. Crucially, the model can discern various object classes and their spatial occupancy, thereby enabling the precise assignment of pre-constructed models to their respective positions in the VR environment.

By doing so, the project seeks to enhance both accuracy and efficiency, enabling automated design processes that do not rely solely on human expertise. This potential integration of machine learning holds promise for advancing the field and expanding the applicability of the intelligent space planning platform to a wider range of facility environments. The fundamental idea underlying this research is that VR can extend beyond its traditional applications in training and entertainment. Instead, it has the potential to be valuable in various everyday life scenarios.

ACKNOWLEDGEMENT

We would like to thank the programme committee and reviewers for their valuable feedback and suggestions. We are grateful to the Innovation and Technology Commission, the Government of the Hong Kong Special Administrative Region of the People's Republic of China, for supporting our research on virtual reality (e.g., through the ITF PSTS project ITT/019/22LP). We would also like to acknowledge the support from the Tam Wing Fan Innovation Fund and Philomathia Foundation Innovation Fund. Finally, we thank Dr. Tobey H. Ko for her technical guidance and support, as well as Professor Henry Lau for his great efforts in establishing the imseCAVE.

REFERENCES

- [1] S. Safikhani, S. Keller, G. Schweiger, and J. Pirker, "Immersive virtual reality for extending the potential of building information modeling in architecture, engineering, and construction sector:

- systematic review," *International Journal of Digital Earth*, vol. 15, no. 1, pp. 503-526, 2022, doi: 10.1080/17538947.2022.2038291.
- [2] M. Slater and M. V. Sanchez-Vives, "Enhancing Our Lives with Immersive Virtual Reality," *Frontiers in Robotics and AI*, vol. 3, 2016, doi: 10.3389/frobt.2016.00074.
- [3] Ş. A. Oğuzcan Ergün, İpek Gürsel Dino, Elif Sürer, "Architectural Design in Virtual Reality and Mixed Reality Environments: A Comparative Analysis," presented at the 2019 IEEE Conference on Virtual Reality and 3D User Interfaces, Osaka, Japan, 2019. [Online]. Available: <https://ieeexplore.ieee.org/document/8798180/>.
- [4] J. Du, Y. Shi, Z. Zou, and D. Zhao, "CoVR: Cloud-Based Multiuser Virtual Reality Headset System for Project Communication of Remote Users," *Journal of Construction Engineering and Management*, vol. 144, no. 2, 2018, doi: 10.1061/(asce)co.1943-7862.0001426.
- [5] M. Weijss-Perrée, J. van de Koevinger, R. Appel-Meulenbroek, and T. Arentze, "Analysing user preferences for co-working space characteristics," *Building Research & Information*, vol. 47, no. 5, pp. 534-548, 2018, doi: 10.1080/09613218.2018.1463750.
- [6] C. Anderson, C. Bailey, A. Heumann, and D. Davis, "Augmented space planning: Using procedural generation to automate desk layouts," *International Journal of Architectural Computing*, vol. 16, no. 2, pp. 164-177, 2018, doi: 10.1177/1478077118778586.
- [7] H. Tanasra, T. Rott Shaham, T. Michaeli, G. Austern, and S. Barath, "Automation in Interior Space Planning: Utilizing Conditional Generative Adversarial Network Models to Create Furniture Layouts," *Buildings*, vol. 13, no. 7, 2023, doi: 10.3390/buildings13071793.
- [8] S. Colenberg, T. Jylhä, and M. Arkesteijn, "The relationship between interior office space and employee health and well-being – a literature review," *Building Research & Information*, vol. 49, no. 3, pp. 352-366, 2020, doi: 10.1080/09613218.2019.1710098.
- [9] J. M. Y. Chim, "6Ws in Ergonomics Workplace Design," in *Proceedings of the 20th Congress of the International Ergonomics Association (IEA 2018)*, (Advances in Intelligent Systems and Computing, 2019, ch. Chapter 129, pp. 1282-1286.
- [10] S. M. Reddy, D. Chakrabarti, and S. Karmakar, "Emotion and interior space design: an ergonomic perspective," *Work*, vol. 41 Suppl 1, pp. 1072-8, 2012, doi: 10.3233/WOR-2012-0284-1072.
- [11] Y. El-Ghazouly and A. El Antably, "Using Digital Human Models to Evaluate the Ergonomic Comfort of Interior Layouts and Furniture Design," *Technology|Architecture + Design*, vol. 5, no. 2, pp. 225-240, 2021, doi: 10.1080/24751448.2021.1967061.
- [12] FARO. "FARO SCENE Software." <https://www.faro.com/en/Products/Software/SCENE-Software> (accessed 2024).
- [13] Votanic. "XR Creation." <https://www.votanic.com/xrcc/> (accessed 2024).
- [14] Autodesk. "Autodesk 3ds Max: Create massive worlds and high-quality designs." <https://www.autodesk.com/products/3ds-max/overview?term=1-YEAR&tab=subscription> (accessed 2023).
- [15] L. Y. Charles R. Qi, Hao Su, Leonidas J. Guibas, "PointNet++: Deep Hierarchical Feature Learning on Point Sets in a Metric Space,"
- [16] Z. Wu, S. Song, A. Khosla, F. Yu, L. Zhang, X. Tang and J. Xiao, "3D ShapeNets: A Deep Representation for Volumetric Shapes," in *2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 7-12 June 2015 2015: IEEE.
- [17] Q.-Y. Z. Jaesik Park, Vladlen Koltun "Colored Point Cloud Registration Revisited," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 143-152.
- [18] K. Schwabe, J. Teizer, and M. König, "Applying rule-based model-checking to construction site layout planning tasks," *Automation in Construction*, vol. 97, pp. 205-219, 2019, doi: 10.1016/j.autcon.2018.10.012.